CLAIMS INTELLIGENCE AGENT POWERED BY COPILOT STUDIO

A Copilot Studio solution integrating risk scoring, hybrid RAG retrieval, and structured AI outputs to enhance claims evaluation efficiency and governance

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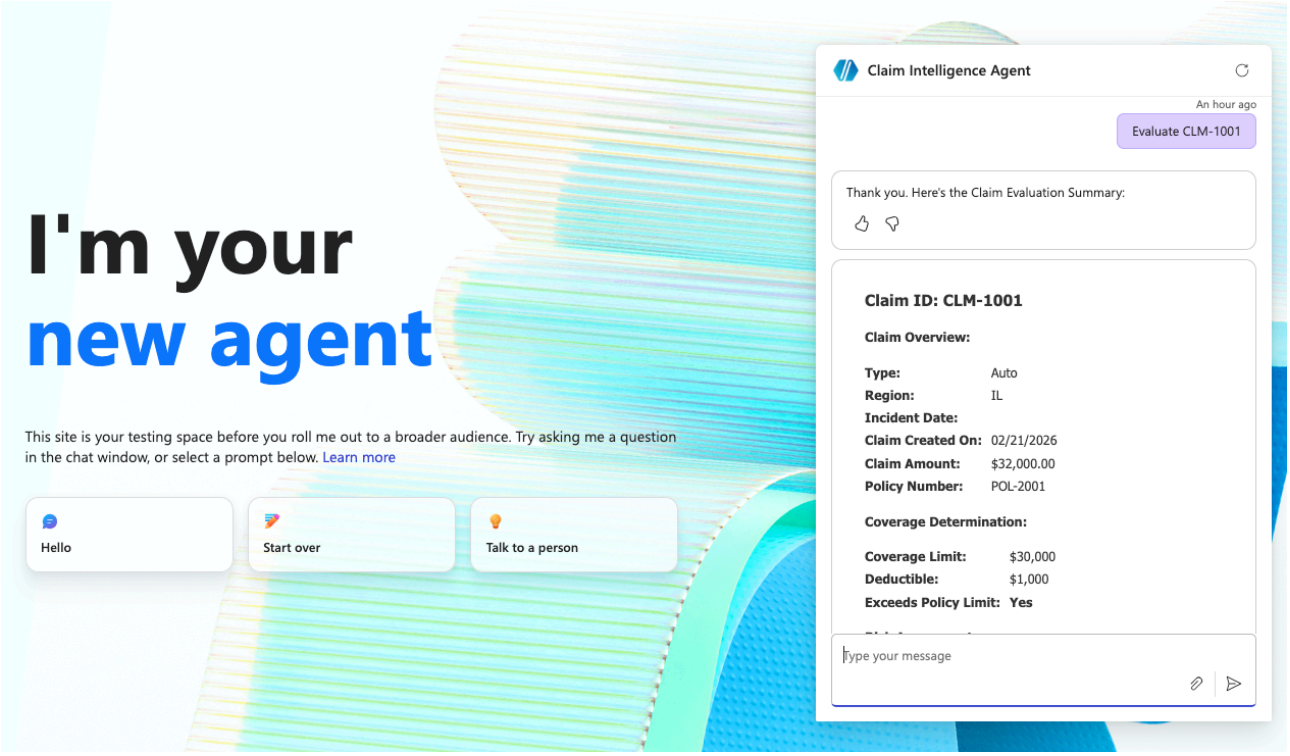


BUSINESS PROBLEM

Claims adjusters need to validate policy coverage and assess risk quickly. Manual document review requires significant time, and contains risks to introduce inconsistency in the business operations and the outcome.

SOLUTION OVERVIEW

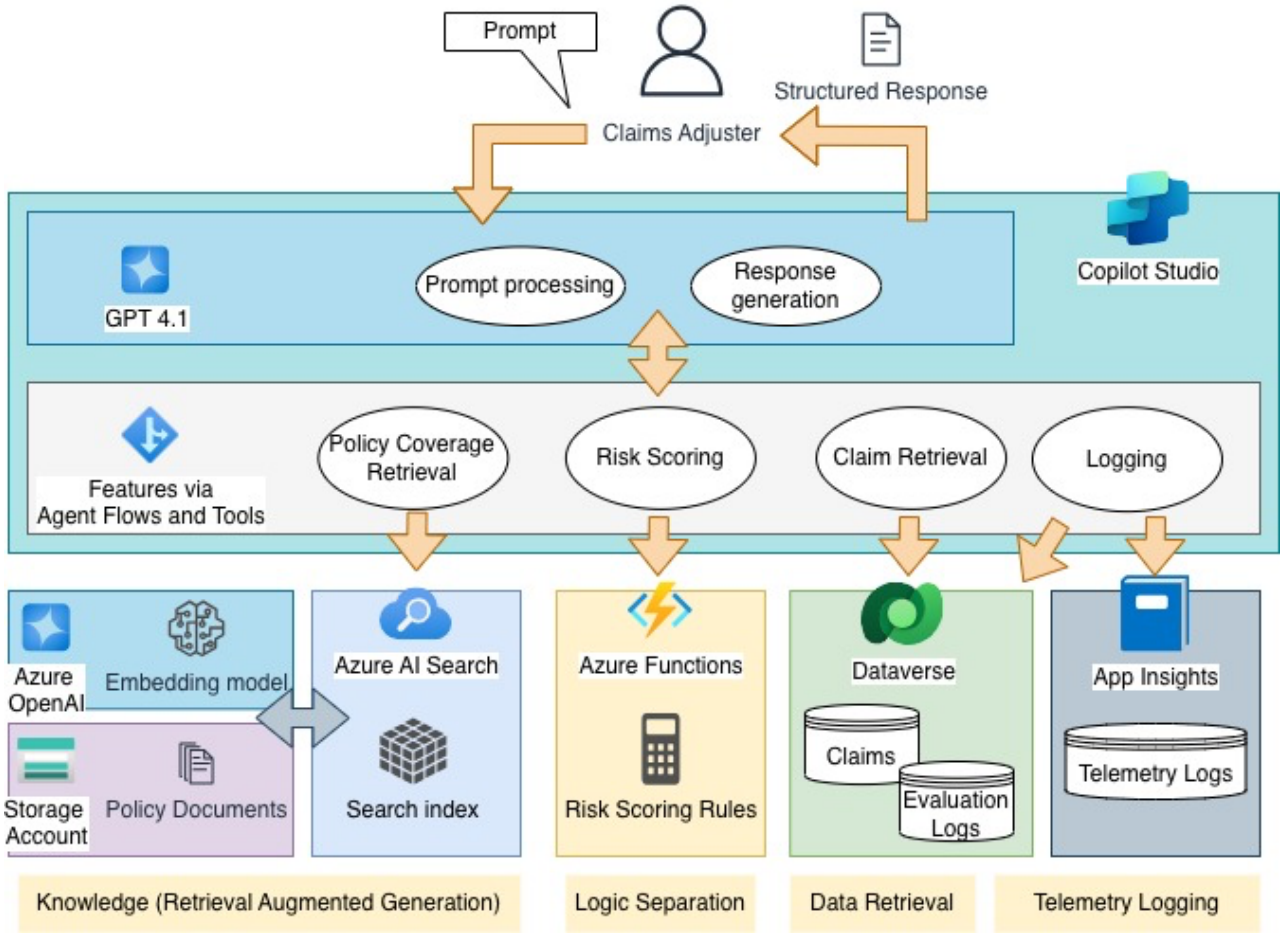
This solution introduces an AI-orchestrated decision support agent designed to enhance claims evaluation accuracy, efficiency and governance alignment.

KEY INNOVATION

- Separation of risk score calculation logic from generative AI solution to ensure reusability, explainability and governance alignment.
- Generation of structured response to ensure consistency and integrity across business users.

ARCHITECTURE OVERVIEW

SYSTEM ARCHITECTURE



SOLUTION COMPONENTS

- Copilot Studio: to develop a custom AI agent to produce interactions using the data and tools residing in Azure and Power Platform
- Azure AI Search: to generate and store search index for Policy data. Azure OpenAI and grounding data storage (Azure Storage Account) are required to produce index.
- Azure OpenAI: to generate vector embeddings for indexing. Indexes are used to generate grounded summaries based on retrieved context.
- Storage Account: to store grounding data (Policy data) in this scenario, sample data are stored in docx, pdf, or txt format.
- Azure Function App: to develop a business logic dedicated to risk scoring and also to be separated from Copilot Studio and Power Platform in terms of reusability and accountability.
- Dataverse: to store insurance claim records generated by clients and business users
- Application Insights: to store telemetry logs for investigation and troubleshooting conducted by system administrators.

ARCHITECTURAL PRINCIPLES

- Retrieves grounded policy information using RAG powered by AI search and Azure OpenAI
- Applies deterministic risk scoring performed in Azure Function App
- Generates structured claim summaries combining data from Dataverse and Storage Account and leveraging Adaptive Card UI framework
- Logs risk score and decisions for auditability using Dataverse and Power Apps

SOLUTION AND SOURCE CODE

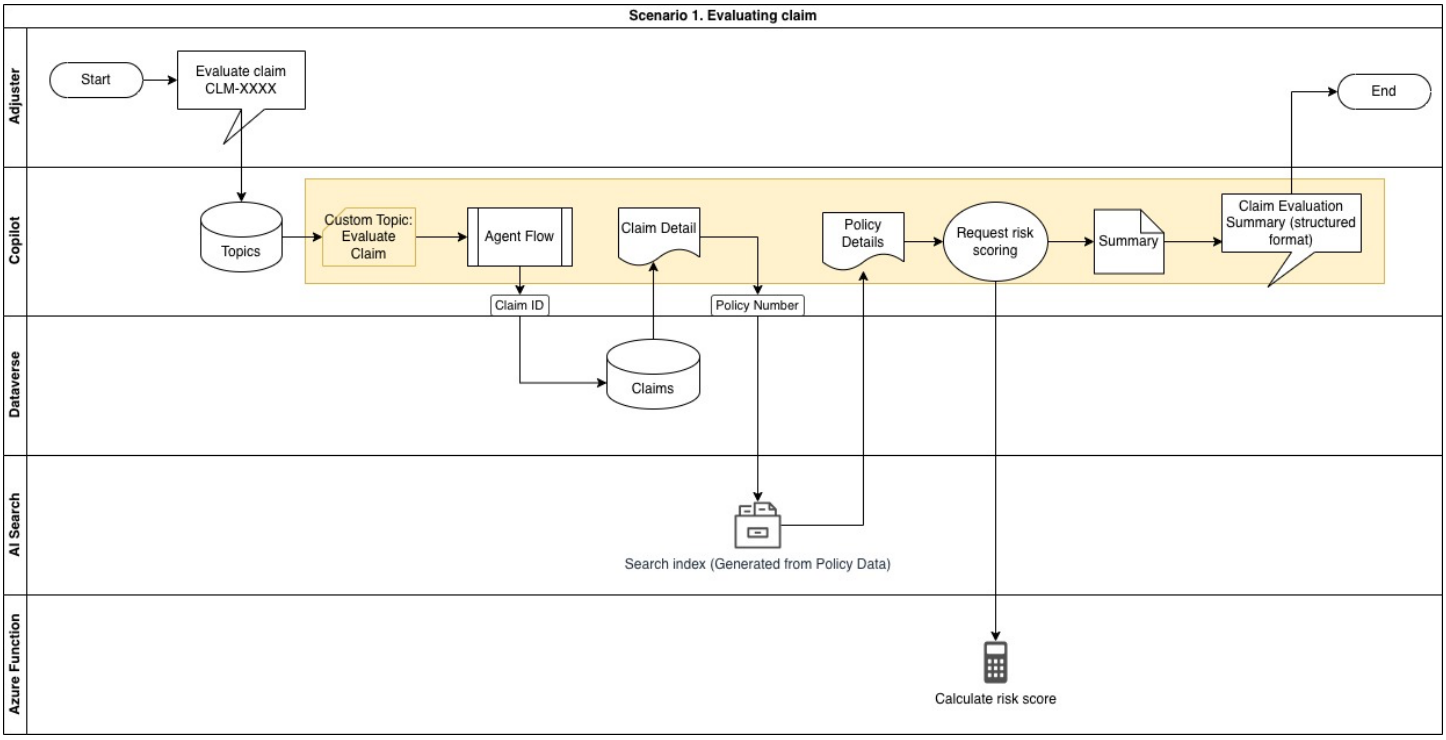
GitHub link: https://github.com/satokora/CopilotClaimIntelligence

BUSINESS SCENARIO

TARGET USER

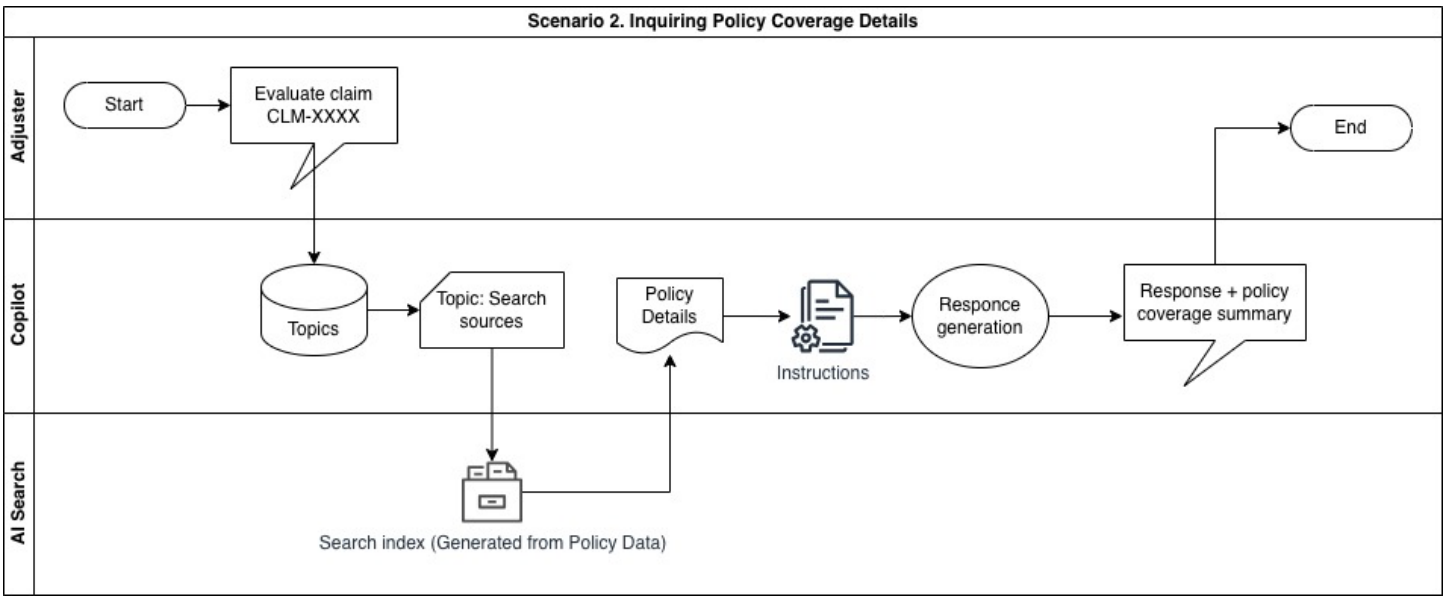
- Claims Adjuster – investigates, evaluates, and settles insurance claims to determines organizational liability and appropriate settlement amounts

WORKFLOW 1: EVALUATING CLAIM



1. Adjuster Submits Claim ID
2. Copilot detects the topic for claim evaluation
3. Copilot retrieves claim details from Dataverse
4. Copilot searches for Policy Coverage details stored in Storage Account via AI search
5. Azure Function processes risk score calculation
6. AI model generates structured summary
7. Risk score results and escalation decision is logged in Dataverse
8. Copilot returns the summary to Adjuster

WORKFLOW 2: INQUIRING POLICY COVERAGE DETAILS



1. Adjuster submits a question with a Policy ID
2. Copilot searches for Policy Coverage details stored in Storage Account via AI search
3. AI model generates a response followed with a structured summary of the policy details
4. Copilot returns the response to Adjuster

TECHNICAL DESIGN DECISIONS

Design Decision	Rationale
Deterministic risk scoring in Azure Function	The logic to calculate risk score based on the claim details is separately implemented in Azure Functions for reusability and explainability
Structured Output (JSON, Adaptive card)	Claim evaluation summary and policy coverage information is formatted in a consistent format for consistency
Hybrid RAG retrieval	Policy data is strictly retrieved only from Policy documents stored in Storage Account to ensure data accuracy and governance

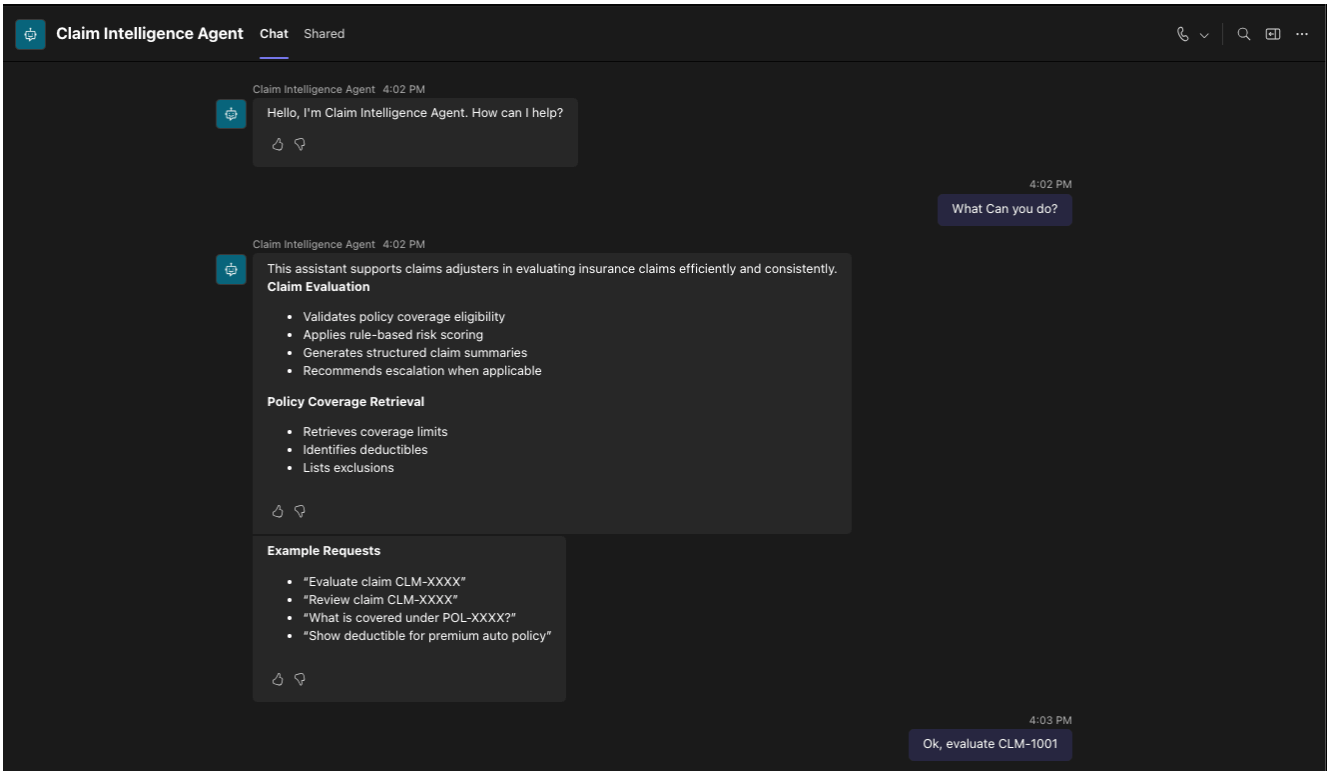
Design Decision	Rationale
Dataverse audit logging	Claim evaluation summary retrieval history is logged in Dataverse for business users' audit purposes
Application Insights Telemetry	Telemetry logging in the agent and the Azure function app is enabled to monitor performance, data accuracy and potential response anomalies.

AI GOVERNANCE & RESPONSIBLE DESIGN

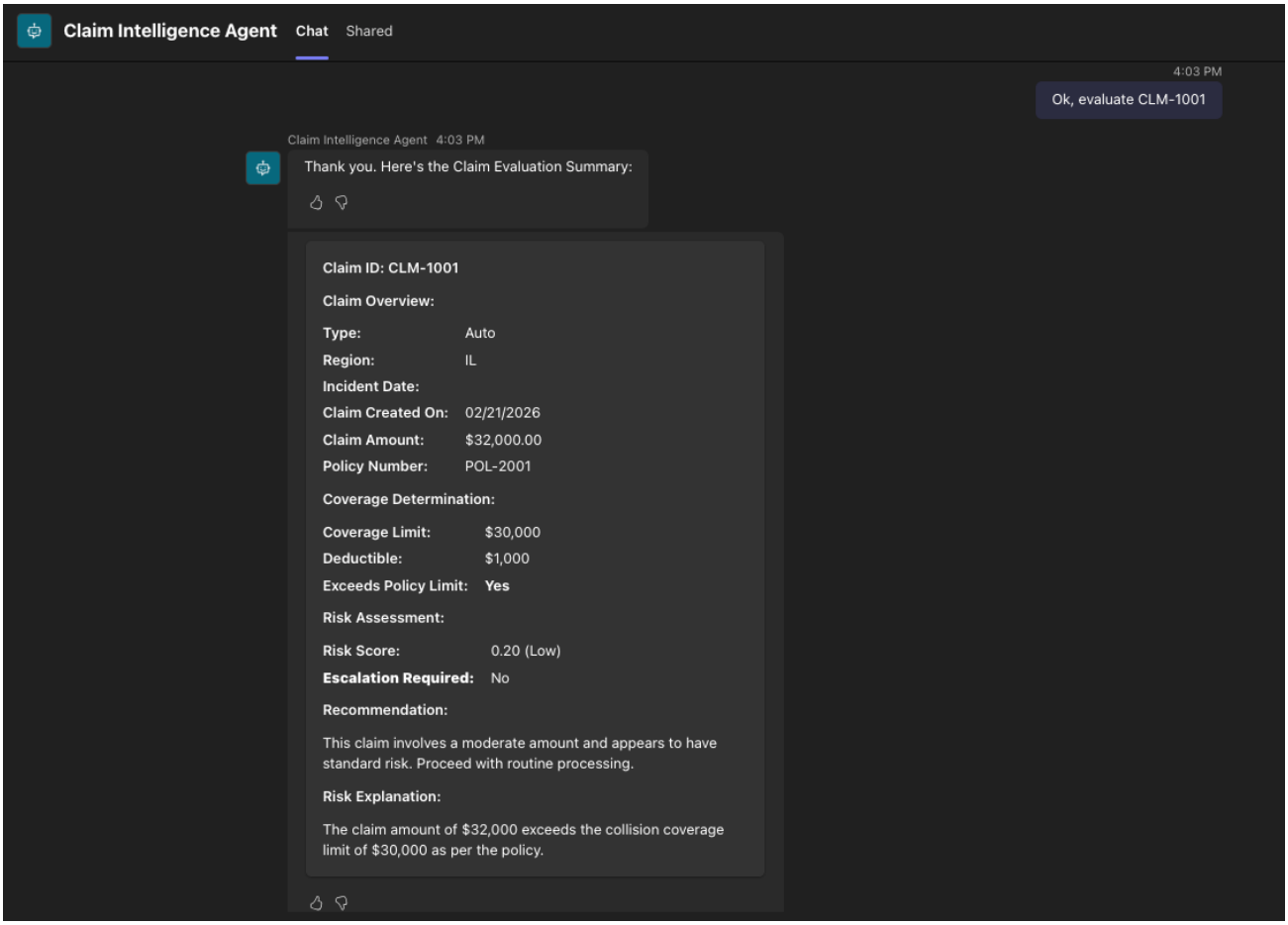
Governance Rule	Rationale
Non-speculation prompt guardrails	Instructions are given to the agent to enforce using only provided grounding data, not fabricating information and explicitly stating information is missing to the users.
Escalation alignment rules	Escalation to the senior adjusters is determined in a logic separated from LLM (Azure Function App in this scenario) to ensure escalation determination rules are always consistent.
Separation of business rules from LLM reasoning	Deterministic risk scoring rules are separately implemented in Azure Function App for accountability and reusability.
Data Minimization	Only structured decision data attributes, such as risk score, escalation flag, timestamp are stored in Dataverse. Full summaries and policy documents are not recorded to avoid unnecessary sensitive data retention exposure and excessive storage usage.

SAMPLE OUTPUT

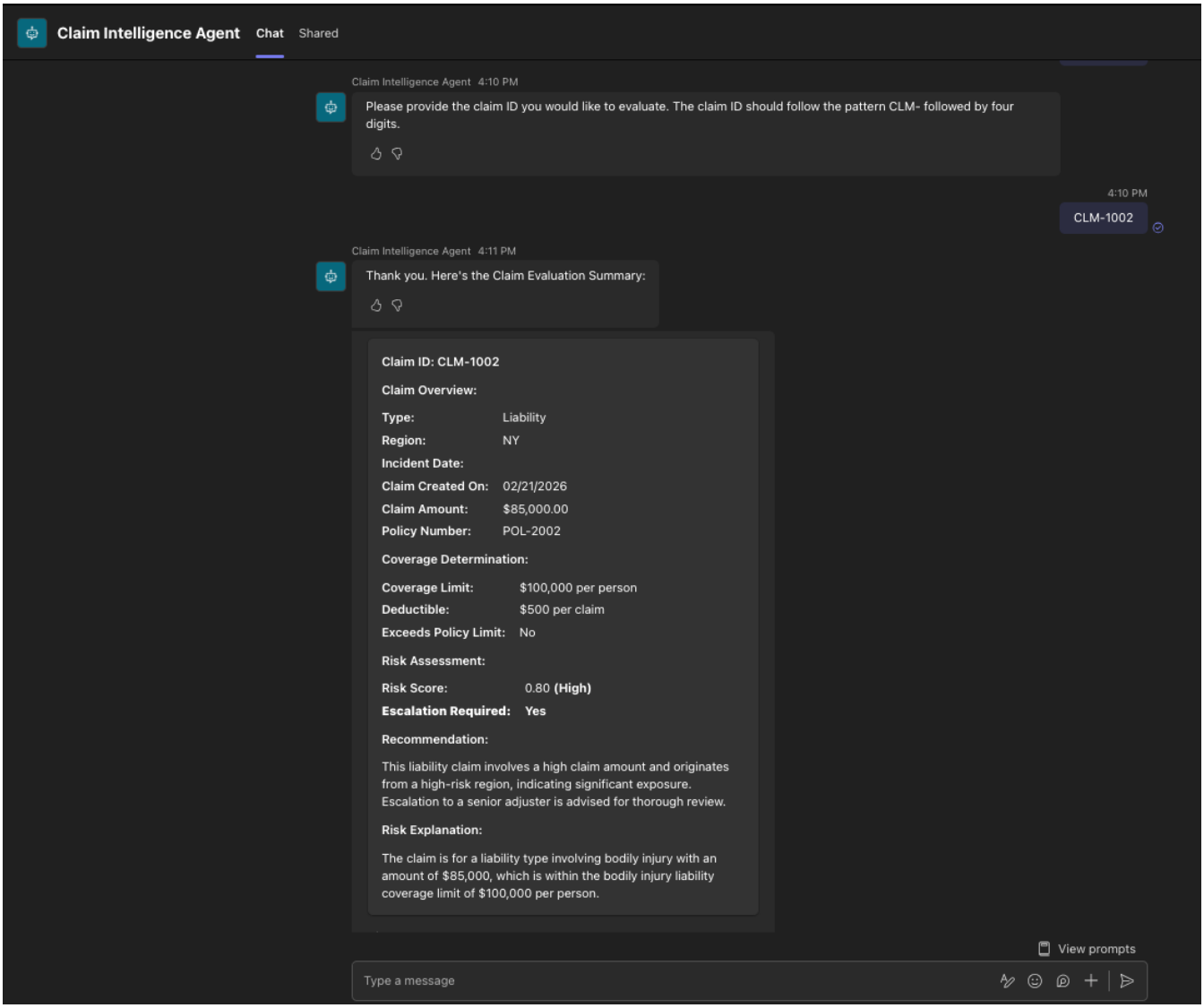
- Beginning of conversation and Agent introduction



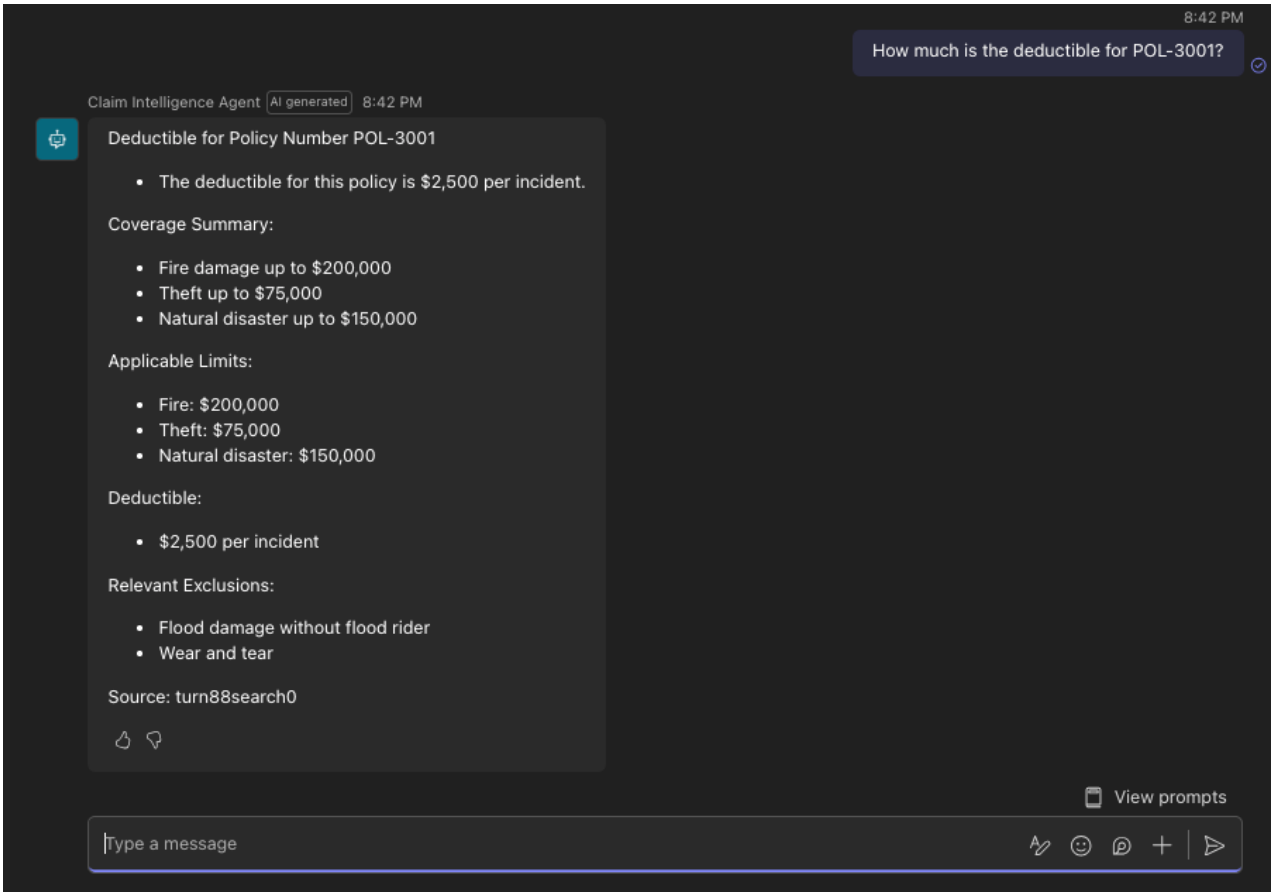
- Low-risk Claim Result



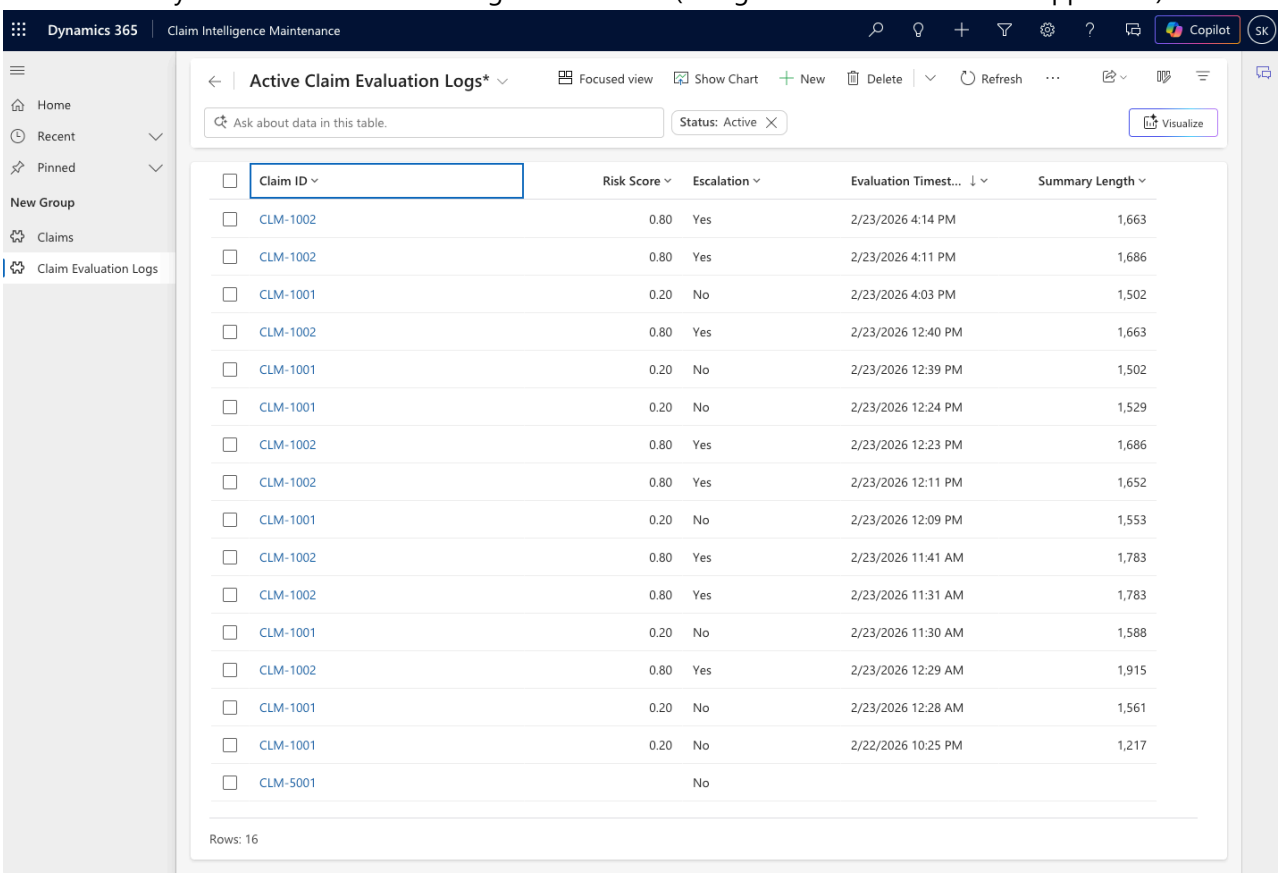
- High-risk Claim Result


Claim Intelligence Agent
Chat
Shared
Claim Intelligence Agent 4:10 PM
Please provide the claim ID you would like to evaluate. The claim ID should follow the pattern CLM- followed by four digits.
4:10 PM
CLM-1002
Claim Intelligence Agent 4:11 PM
Thank you. Here's the Claim Evaluation Summary:
Claim ID: CLM-1002
Claim Overview:
Type: Liability
Region: NY
Incident Date:
Claim Created On: 02/21/2026
Claim Amount: $85,000.00
Policy Number: POL-2002
Coverage Determination:
Coverage Limit: $100,000 per person
Deductible: $500 per claim
Exceeds Policy Limit: No
Risk Assessment:
Risk Score: 0.80 (High)
Escalation Required: Yes
Recommendation:
This liability claim involves a high claim amount and originates from a high-risk region, indicating significant exposure. Escalation to a senior adjuster is advised for thorough review.
Risk Explanation:
The claim is for a liability type involving bodily injury with an amount of $85,000, which is within the bodily injury liability coverage limit of $100,000 per person.
View prompts
Type a message

- Policy Q&A

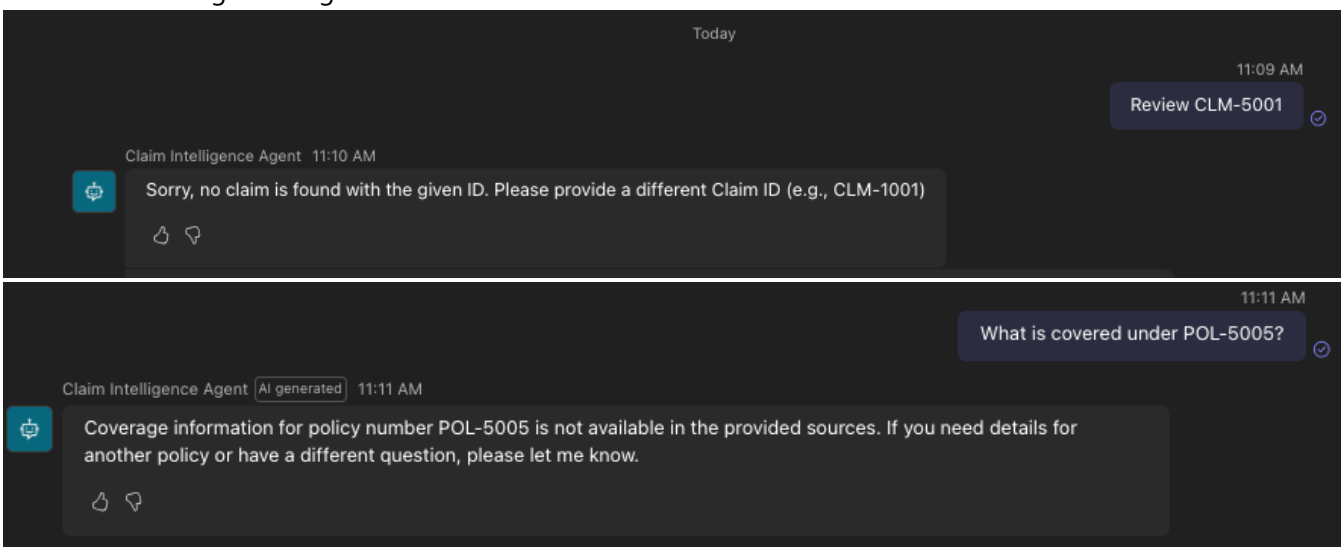


- Claim Summary retrieval and risk score logs in Dataverse (Using a custom Model-driven app for UI)



Claim ID	Risk Score	Escalation	Evaluation Timest...	Summary Length
CLM-1002	0.80	Yes	2/23/2026 4:14 PM	1,663
CLM-1002	0.80	Yes	2/23/2026 4:11 PM	1,686
CLM-1001	0.20	No	2/23/2026 4:03 PM	1,502
CLM-1002	0.80	Yes	2/23/2026 12:40 PM	1,663
CLM-1001	0.20	No	2/23/2026 12:39 PM	1,502
CLM-1001	0.20	No	2/23/2026 12:24 PM	1,529
CLM-1002	0.80	Yes	2/23/2026 12:23 PM	1,686
CLM-1002	0.80	Yes	2/23/2026 12:11 PM	1,652
CLM-1001	0.20	No	2/23/2026 12:09 PM	1,553
CLM-1002	0.80	Yes	2/23/2026 11:41 AM	1,783
CLM-1002	0.80	Yes	2/23/2026 11:31 AM	1,783
CLM-1001	0.20	No	2/23/2026 11:30 AM	1,588
CLM-1002	0.80	Yes	2/23/2026 12:29 AM	1,915
CLM-1001	0.20	No	2/23/2026 12:28 AM	1,561
CLM-1001	0.20	No	2/22/2026 10:25 PM	1,217
CLM-5001		No		

- When there is no grounding data



FUTURE ENHANCEMENTS

Future Enhancement	Details
Agent-to-agent modular copilots	Separating tasks in multiple agents and implementing a parent agent in charge of orchestration reduces bottlenecks and improves accuracy.
Entra ID authentication + private endpoints	Enforcing Entra ID authentication for access control and applying additional security layers such as private endpoints to Azure resources such as Azure Function App for robust, enterprise-level security.
Increased availability and performance optimization	Upgrading service plan for Azure Function App, more robust Policy data distribution across multiple sources (Dataverse and Storage Account for instance), and Advanced business logic distribution balancing between Topic, Agent flows, and Azure Function app could contribute to increased availability and improved performance.
Dashboard analysis for operational insights	Leveraging Application Insights, Dataverse and Power BI integration enables to develop an analytical dashboard for operational insights and data monitoring.

CLOSING STATEMENT

This solution demonstrates enterprise AI orchestration, governance-aware design, and modular architecture aligned with modern Copilot Studio development practices.